Software Project Management
Assignment-6
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following statements regarding a Work Breakdown Structure (WBS) are correct?

a. A WBS represents a hierarchical decomposition of project work.
b. A WBS decomposes tasks into smaller subtasks.
c. A WBS shows the duration of each activity.
d. A WBS helps organize project scope into manageable components.
e. A WBS explicitly shows task dependencies.

Correct Answer: a. A WBS represents a hierarchical decomposition of project work.
b. A WBS decomposes tasks into smaller subtasks.
d. A WBS helps organize project scope into manageable components

Detailed Solution:

A WBS is a hierarchical decomposition of the total project work into smaller, manageable components (work packages), which helps organize and define the project scope. It decomposes tasks into subtasks but does not show activity durations (that belongs to schedules like Gantt charts) or task dependencies (shown in network diagrams).

QUESTION 2:

Which of the following statements regarding an Activity Network are correct?

a. It cannot represent concurrent activities.
b. An Activity Network depicts the flow of project tasks.
c. It is primarily used to define software requirements.
d. It can show task dependencies.
e. It can represent task sequence relationships.

Correct Answer: b. An Activity Network depicts the flow of project tasks.
d. It can show task dependencies.
e. It can represent task sequence relationships

Detailed Solution:

An Activity Network (e.g., PERT or CPM) depicts the logical flow and sequence of project tasks, clearly showing task dependencies and relationships (e.g., finish-to-start). It can also represent concurrent activities using parallel paths, so option (a) is false. It is used for project scheduling, not for defining software requirements, making option (c) incorrect.

QUESTION 3:

Which of the following pieces of information is typically represented in a WBS?

a. Major project deliverables.
b. Critical path information.
c. Subtasks associated with project activities.
d. Activity durations.
e. Hierarchical relationships among work items.

Correct Answer: a. Major project deliverables
c. Subtasks associated with project activities
e. Hierarchical relationships among work items

Detailed Solution:

A WBS typically represents major project deliverables, subtasks (work packages) associated with project activities, and hierarchical relationships among work items. It does not include critical path information (shown in network diagrams) or activity durations (shown in schedules like Gantt charts).

QUESTION 4:

Which statements about the critical path are true?

a. It always includes the maximum number of activities.
b. It is a path from project start to finish.
c. It determines the minimum project duration.
d. It always contains all project tasks.
e. Delays on critical activities may delay the project.

Correct Answer: b. It is a path from project start to finish.
c. It determines the minimum project duration
e. Delays on critical activities may delay the project

Detailed Solution:

The critical path is the longest sequence of dependent activities from project start to finish, and it determines the minimum project duration. Delaying any critical activity will delay the project unless corrective action is taken. It does not necessarily contain the maximum number of activities (duration matters more), and it never includes all project tasks—only those with zero slack.

QUESTION 5:

What can be the effect of the delay of an activity in the critical path?

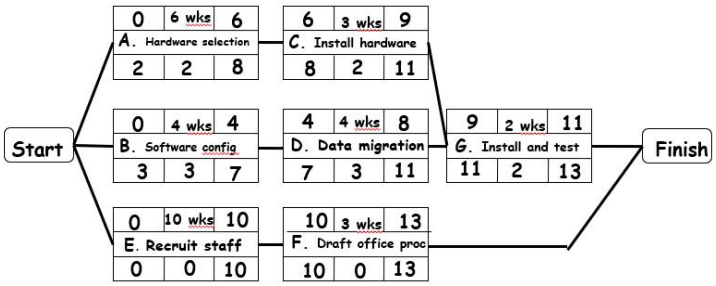
a. The project completion date may be delayed.
b. The critical path may change.
c. The project duration is guaranteed to remain unchanged.
d. The delay is always absorbed by slack time.
e. Corrective actions may be required.

Correct Answer: a. The project completion date may be delayed.
b. The critical path may change.
e. Corrective actions may be required

Detailed Solution:

Delaying a critical-path activity directly threatens the project completion date (a), may cause the critical path to shift to another sequence of activities (b), and typically forces the project manager to take corrective actions such as crashing or fast-tracking to recover (e). Critical activities have zero slack, so the delay is never absorbed by slack time (d), and the project duration is not guaranteed to remain unchanged (c).

QUESTION 6:



Which activities have zero float (slack) in the network in the above PERT chart?

a. A
b. B
c. E (Recruit staff)
d. F (Draft office procedures)
e. G (Install and test)

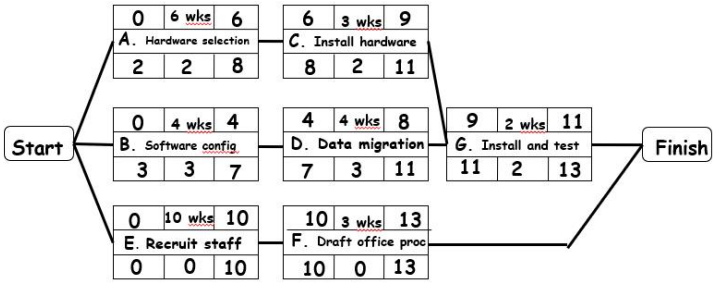
Correct Answer: c. E (Recruit staff)
d. F (Draft office procedures)

Detailed Solution:

E (Recruit staff)
ES=0, DUR=10, EF=10
LS=0, SL=0, LF=10

F (Draft office procedures)
ES=10, DUR=3, EF=13
LS=10, SL=0, LF=13

QUESTION 7:



Which of the following activities are on the critical path in the given PERT chart diagram?

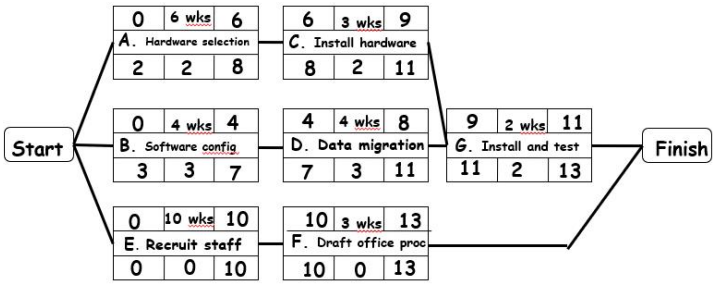
a. A
b. E (Recruit staff)
c. F (Draft office procedures)
d. C
e. G

Correct Answer: b. E (Recruit staff)
c. F (Draft office procedures)

Detailed Solution:

The critical path is the longest path from start to finish with zero slack: E (Recruit staff) → F (Draft office procedures). No other activities are on the critical path because they have positive slack and are not required to be completed on the critical timeline.
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QUESTION 8:



Suppose activity D is delayed by 2 weeks while all other activities remain unchanged in the above PERT chart diagram. Which statements are true?

a. Activity G must start at week 11.
b. D will finish at week 10.
c. The project completion time remains 13 weeks.
d. D will still not be on the critical path.
e. The float of D becomes 1 week.

Correct Answer: b. D will finish at week 10.
c. The project completion time remains 13 weeks.
d. D will still not be on the critical path

Detailed Solution:

D will finish at 10 (8+2) where the completion will remain 13 as both EF =LF=13, and D will still not be on the critical path.

QUESTION 9:

Which statements about CPM/PERT are correct?

a. Activities with larger durations always have larger float.
b. Forward pass computes earliest times.
c. All activities can have non-zero float.
d. Critical-path activities have positive float.
e. Backward pass computes latest times.
f. Float = LS – ES.

Correct Answer: b. Forward pass computes earliest times.
e. Backward pass computes latest times.
f. Float = LS – ES.

Detailed Solution:

In CPM/PERT, the forward pass calculates earliest start and finish times (b), while the backward pass calculates latest start and finish times (e). Float (slack) is correctly computed as LS – ES (or LF – EF) (f). Larger duration does not guarantee larger float (a), critical activities have zero float (d), and not all activities can have non-zero float—critical ones have zero (c).

QUESTION 10:

Given an activity with Earliest Start (ES) = 5 days, Latest Finish (LF) = 30 days, and Duration = 10 days, which of the following are true?

a. Earliest Finish (EF) = 15 days
b. Float = 20 days
c. Latest Start (LS) = 20 days
d. Float = 15 days
e. Latest Start (LS) = 25 day

Correct Answer: a. Earliest Finish (EF) = 15 days
c. Latest Start (LS) = 20 days
d. Float = 15 days

Detailed Solution:

EF = ES + Duration = 5 + 10 = 15 days → (a) is true.
LS = LF – Duration = 30 – 10 = 20 days → (c) is true, (e) is false.
Float = LS – ES = 20 – 5 = 15 days → (d) is true, (b) is false.
Thus, the correct statements are a, c, and d.

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